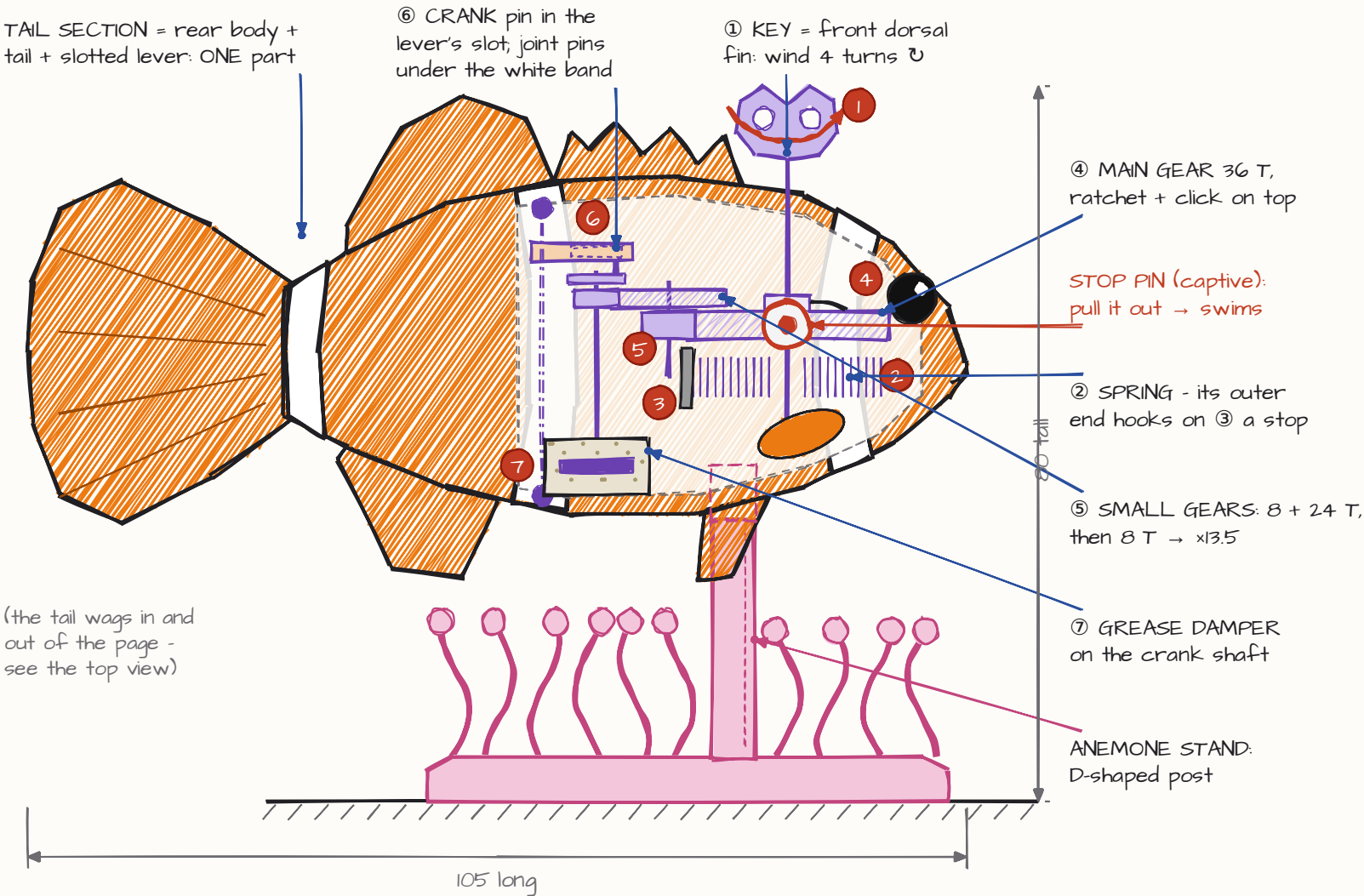


CLOWNFISH WIND-UP - HOW IT WORKS

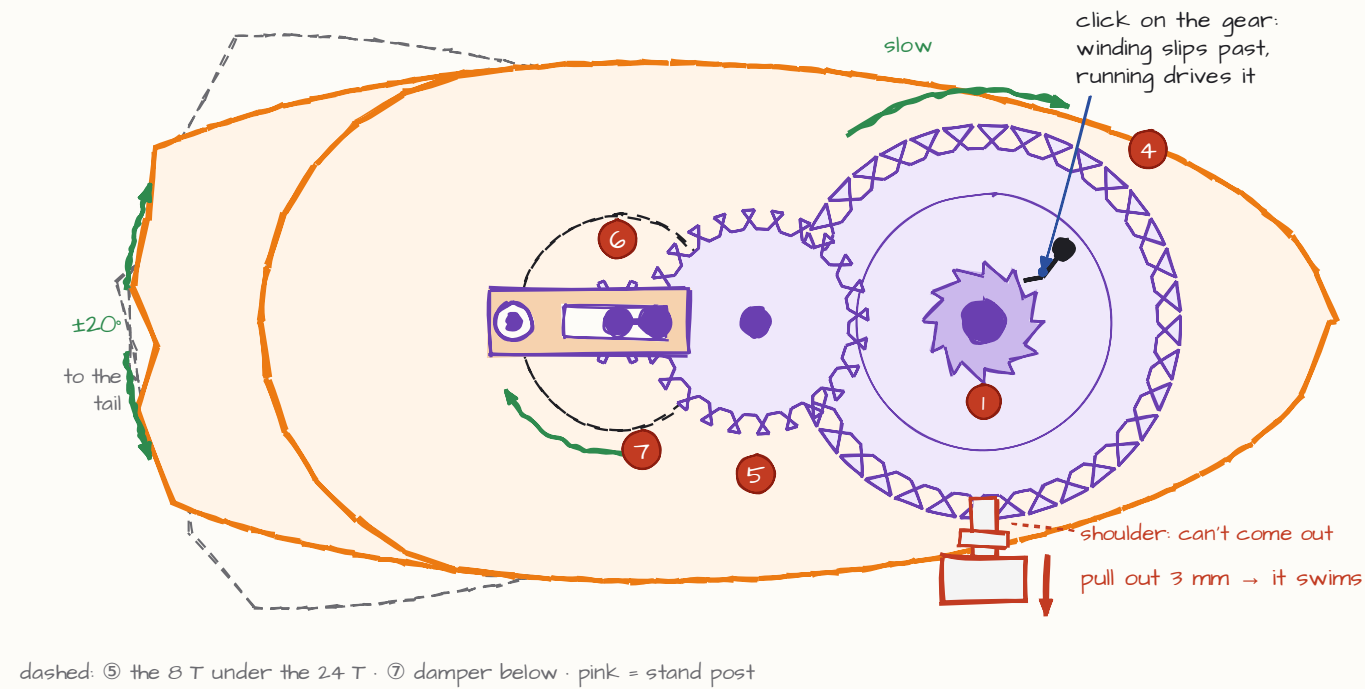
key → spring → main gear → small gears → crank → the tail wags in place · purple = mechanism

SIDE VIEW - cut away · from the fish's right · scale 1.4 : 1

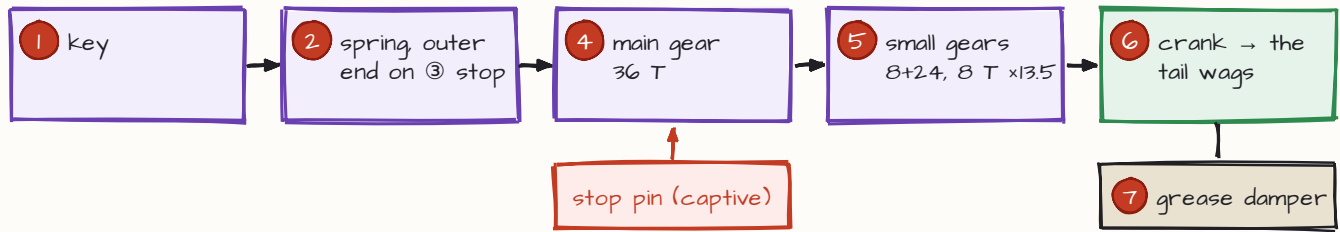


TOP VIEW - CUT THROUGH THE GEARS

seen from above · scale 2.3 : 1



SAME MOTOR AS THE WIND-UP CAR - PLUS 3 EXTRAS



From your picture: the key ① winds the spring ②; a stop ③ holds its outer end. The spring turns the main gear ④, which spins the small gear ⑤ fast. In the car ⑤ turns the wheels; here it turns a crank ⑥.

What the fish adds:

- a click, so winding doesn't drive the gears
- a 2nd small-gear stage → 3x more wags
- ⑦ a grease damper: the car spends its energy rolling a fish standing still would spin out in under 1/2 s

HOW IT PLAYS

- 1 Push the stop pin IN - it drops between the main-gear teeth
- 2 Wind the dorsal-fin Key 4 turns.
- 3 Set the fish on its anemone stand.
- 4 Pull the stop pin OUT → the tail wags 2-3x a second, still $\geq 1\times$ a second after 15 s.
- 5 Push the pin back in to stop.

The key turns slowly backwards while it swims - don't hold it.

WHY IT LASTS ≥ 5 s

- Spring: PETG strip 0.45×5 , ≈ 400 mm (11 coils), wound 4 turns → ≈ 5 Nmm, ≈ 60 mJ, 30 MPa (stop at 4 turns)
- Gears $\times 13.5$ (36:8 then 24:8): up to 54 crank turns = 54 wags (≈ 45 used)
- No damper → it races, empty in $< 1/2$ s
- ⑦ Ø8 rotor, 2 mm gaps of 30 Pas silicone grease → $c \approx 0.012$ Nmm/s
- Wag rate = (crank torque - friction) $\div 2\pi c$: 2.6/s at the start, 2.0/s at 5 s, $\geq 1/s$ for 16 s
- Damper $\times 2$ or ≈ 2 , double friction or a weaker spring: still $\geq 1/s$ for ≥ 9 s

PARTS (10) + RULES

- 1-2 head halves (snap together)
3 key + arbor + ratchet + spiral spring
5 main gear 36 T + click 6 gear 8 + 24 T
7 crank pinion 8 T + crank + damper rotor
8 tail section (body, tail, slotted lever)
9 stop pin (captive) 10 anemone stand

- ✓ All printed + a dab of damping grease
- ✓ Fits the egg; built 105×80 → bigger
- ✓ Stop pin is captive: it pulls out but can't come off (no removable pin)
- ✓ Opening the head lets the spring go
- ✓ Snap fits: no glue, no tools

BIOMIMICRY

Clownfish don't cruise: they hover beside their anemone, sculling with quick tail beats. This one does the same on its anemone stand; its middle white band hides the tail joint.

ENGG*2100 F26 · Design & Build · Concept Sketch

Concept 4 · sheet 1 of 1

Concept 4: CLOWNFISH WIND-UP

Theme: Biomimicry - clownfish hovering at its anemone
Toy type: Spring wind-up, swims in place ≥ 5 s
Scale / units: mm · side view 1.4 : 1 on 11x17

Drawn by: _____ Signature: _____ Date: _____